1. Primary Alignment: Items 9 and 5

- **Item 9: Understand the relationship between the context of the problem and representation of the mathematical solution...** (pp. 1-2)
  - **Analysis:** This is the core competency tested. Students must bridge the gap between the real-world context (payment rules) and the mathematical representation (lines and piecewise functions in a coordinate system). They must translate verbal descriptions of fixed bases and variable rates into slopes and y-intercepts to interpret the solution in context. (pp. 1-2)
- **Item 5: Interpret mathematical outcomes in a variety of formats... compare or evaluate two or more representations in relation to a situation** (p. 2)
  - **Analysis:** The question presents four different graphical formats (A, B, C, and D) and requires students to evaluate which representation accurately reflects the situation. This involves comparing how the same mathematical relationships are depicted across different visual models. (p. 2)

2. Significant Alignment: Items 1 and 6

- **Item 1: Interpret a mathematical result back into the real world context** (pp. 1-2)
  - **Analysis:** Students must look at the graphs (mathematical results) and interpret their real-world meaning—for instance, identifying that Zedland Daily starts with a higher base pay (60 zeds) while Zedland Star's income grows faster (higher slope) after a certain sales volume. (pp. 1-2)
- **Item 6: Use knowledge of how the real world impacts the outcomes and calculations... to make contextual judgments** (p. 1)
  - **Analysis:** Students use their understanding of how real-world pay works (e.g., "base pay" corresponds to a y-intercept; "commission" corresponds to a slope; "bonus after 240 papers" corresponds to a change in slope/kink in the line) to make a judgment about which graph is correct. (p. 1)

3. Secondary/Indirect Alignment: Items 7 and 8

- **Item 7: Construct and communicate explanations and arguments in the context of the problem** (p. 2)
  - **Analysis:** Although this is a multiple-choice question, the process of selection requires the student to internally construct a logical argument (e.g., "Graph B is incorrect because the line for Zedland Star should change steepness at 240 papers, not remain constant"). (p. 2)
- **Item 8: Recognise (demonstrate, interpret, explain) the extent and limits of mathematical concepts and mathematical solutions** (p. 2)
  - **Analysis:** The task requires identifying the correct model. Students must recognize the limits of certain graphs—for example, noticing that some fail to account for the specific threshold where Zedland Star's pay rate doubles. (p. 2)

1. 核心关联：第 9 项与 第 5 项

- **第 9 项：Understand the relationship between the context of the problem and representation of the mathematical solution...**
  - **分析：** 这是本题最核心的考点。学生必须理解现实背景（两家报纸的付酬规则）与数学表达（坐标系中的直线与折线）之间的关系 (pp. 1-2)。学生需要将文字描述的线性增长和分段函数转化为图形上的斜率和截距，从而判断方案在不同销售量下的可行性。
- **第 5 项：Interpret mathematical outcomes in a variety of formats... compare or evaluate two or more representations in relation to a situation**
  - **分析：** 题目提供了四种不同的图形表示（A、B、C、D），要求学生评估哪一个能正确对应现实情况 (p. 2)。这涉及对同一数学关系在不同图形呈现下的对比与评价。

2. 重要关联：第 1 项与 第 6 项

- **第 1 项：Interpret a mathematical result back into the real world context**
  - **分析：** 学生需要通过观察图形（数学结果），将其解读回现实意义 (p. 2)。例如，Zedland Daily 的起始收入较高（60 zeds），而 Zedland Star 的收入在销售量达到一定程度后增长更快（斜率变大）。
- **第 6 项：Use knowledge of how the real world impacts the outcomes and calculations... to make contextual judgments**
  - **分析：** 学生需运用对现实世界报酬模式的理解（如"底薪"对应纵轴截距，"提成"对应斜率，"销量增加后提成翻倍"对应折线转折点）来做出判断 (p. 1)。

3. 次要/间接关联：第 7 项与 第 8 项

- **第 7 项：Construct and communicate explanations and arguments in the context of the problem**
  - **分析：** 虽然此题是多选题（Circle A, B, C or D），但在解题思考过程中，学生必须在内心构建一套逻辑解释（例如：排除 B 是因为 Zedland Star 的斜率应该在中途变大，而不是一直恒定）来支持自己的选择 (p. 2)。
- **第 8 项：Recognise... the extent and limits of mathematical concepts and mathematical solutions**
  - **分析：** 题目要求识别正确模型。学生必须意识到不同模型的局限性，例如某些图形未能正确表达 Zedland Star 在 240 份之后报酬增加这一关键转折 (pp. 1-2)。